

SHAGUN TYAGI

AI/ML Engineer • LLM Agents & ML Pipelines • B.Tech CSE + MBA (Expected 2027)

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PROFESSIONAL SUMMARY

AI/ML Engineer with 1+ year of production experience building LLM agents, RAG systems, ML pipelines, and FastAPI microservices at scale. Currently architecting AI-driven SEO intelligence systems at Envigo (LangChain · AWS · Azure · Docker). Previously at Airtel, shipped FastAPI microservices cutting API latency by 30% and Python automation eliminating 10+ hours/week of manual ops. Portfolio spans OpenAI-powered applications, browser automation, computer vision, IoT health monitoring, and full-stack AI products. Published researcher in ECG/PPG-based health monitoring. Pursuing an MBA to combine engineering depth with business-outcomes thinking.

TECHNICAL SKILLS

Languages & Frameworks: Python (FastAPI, Flask, LangChain, TensorFlow, OpenCV, Pandas, NumPy, Scikit-Learn), JavaScript, C

AI / ML / GenAI: LLM Agents, RAG Pipelines, OpenAI API, Prompt Engineering, NLP, Computer Vision, FAISS, ChromaDB, DeepFace, ArcFace, Speech Recognition

Automation & Scraping: Playwright, Browser Automation, Web Crawling, Lead Generation Pipelines, Email Automation

Cloud & DevOps: AWS (EC2, S3, Lambda), Azure (AZ-900, PL-900), Docker, CI/CD, Linux

Data & Databases: SQL, MySQL, MongoDB, SQLite, Vector Databases, Power BI, ETL Pipelines

Frontend & IoT: React 19, Vite, Three.js, Framer Motion, Firebase, React Native, Twilio, Raspberry Pi, GitHub Pages

PROFESSIONAL EXPERIENCE

AI/ML Engineer

Aug 2025 – Present

Envigo — Gurugram, India

- Architect end-to-end ML pipelines on AWS and Azure powering Envigo's SEO intelligence platform — ingesting client data, running AI-driven analysis, and surfacing actionable recommendations at scale.
- Built LangChain-based LLM agents integrated into backend APIs, reducing manual data processing by ~40% across core business workflows.
- Own the full ML lifecycle: model training → Docker containerization → cloud deployment → monitoring and continuous optimization.
- Collaborate with product and data teams to translate ML output into measurable business KPIs, not just accuracy benchmarks.
- Stack: Python · LangChain · FastAPI · AWS · Azure · Docker

Software Engineer

Apr 2025 – Aug 2025

Airtel — Gurugram, India

- Designed and deployed FastAPI microservices powering Airtel's internal ops workflows, reducing API response latency by ~30%.
- Built Python automation pipelines eliminating ~10 hours/week of manual tasks across the operations team.
- Engineered real-time data pipelines feeding dashboards used daily by analytics and ML teams.
- Optimized backend query performance and system architecture, improving throughput by ~25% within 3 months.
- Stack: FastAPI · Python · SQL · Pandas · REST APIs · CI/CD

PROJECTS

PitchIQ — AI Website Auditing & Lead Intelligence Platform | [\[GitHub\]](#)

Stack: Python · OpenAI · Playwright · MongoDB · Email Automation

- Engineered a full-stack AI platform that crawls any website end-to-end and runs automated audits across SEO health, Core Web Vitals, accessibility compliance, UI/UX quality, and content scoring — generating a structured, actionable report in minutes instead of hours of manual consulting.
- Integrated OpenAI GPT to synthesize raw audit data into plain-English recommendations tailored to the business type, enabling non-technical owners to act on findings immediately.
- Built an automated email delivery pipeline that sends personalized audit reports directly to business owners, turning the tool into a self-running lead generation engine for consulting outreach.
- Architected with MongoDB for persistent report storage, enabling trend tracking and comparison across multiple audit runs per client.

LeadGen AI — Google Maps Lead Generation & Outreach Engine | [\[GitHub\]](#)

Stack: Python · OpenAI · Playwright · MongoDB · SMTP Automation

- Built an end-to-end automated lead generation system that scrapes Google Maps for targeted businesses by niche and location, extracts website URLs, phone numbers, and owner contact details using Playwright-driven browser automation.
- Developed an AI enrichment layer using OpenAI GPT that analyzes each lead's website to generate a personalized outreach email referencing their specific business pain points — replacing generic cold email blasts with contextual, conversion-optimized messaging.
- Implemented a lead scoring module that qualifies prospects based on website quality, online presence, and inferred business size — prioritizing high-value targets before outreach.
- Stored all lead data in MongoDB with CSV export support; full pipeline runs autonomously from search query to sent email with zero manual intervention.

Enterprise RAG Chatbot — Multi-Document AI Knowledge System | [\[GitHub\]](#)

Stack: Python · OpenAI · LangChain · FAISS · ChromaDB · MongoDB

- Architected a production-grade Retrieval-Augmented Generation system that ingests PDFs, Word documents, SOPs, and internal manuals — chunking, embedding, and indexing them into a FAISS/ChromaDB vector store for millisecond semantic retrieval.
- Built a LangChain-powered QA pipeline with citation-backed answers that references the exact source document and page, eliminating hallucinations and giving users verifiable, trustworthy responses.
- Engineered context preservation across multi-turn conversations using LangChain's ConversationBufferWindowMemory, enabling follow-up questions without losing thread — behaving like a knowledgeable colleague, not a search box.
- Designed a knowledge base creation workflow allowing non-technical users to upload documents and have a queryable AI assistant operational in under 60 seconds.

FaceGuard — Real-Time AI Facial Recognition Attendance System | [\[GitHub\]](#)

Stack: Python · OpenCV · DeepFace · ArcFace · SQLite · Tkinter

- Built a real-time facial recognition attendance system using ArcFace embeddings — one of the highest-accuracy face recognition architectures — achieving sub-second identification even across varied lighting conditions and partial occlusions.
- Implemented multi-threaded video processing with OpenCV to handle live camera feeds without frame drops, running recognition and UI rendering on separate threads for smooth real-time performance.
- Designed automatic Present/Late/Absent classification based on configurable time thresholds, with all attendance records persisted in SQLite and exportable to structured reports.
- Delivered a desktop dashboard (Tkinter) for administrators to register new faces, view daily attendance logs, generate reports, and manage the system — no command-line knowledge required.

IoT Health Monitoring System — Published Research | [\[GitHub\]](#)

Stack: Python · Raspberry Pi · Scikit-Learn · Firebase · React Native

- Designed and built a wearable-grade IoT health monitor using Raspberry Pi that continuously captures heart rate and SpO2 (blood oxygen) levels via PPG sensors, processing raw physiological signals in real time with Python signal processing pipelines.
- Trained Scikit-Learn classification models on collected health data to detect early indicators of cardiac stress and oxygen desaturation, enabling proactive health alerts before conditions become critical.
- Engineered a Firebase-backed real-time alert system integrated with a React Native mobile app that pushes emergency notifications to caregivers and family members when anomalous readings are detected.

- Research published in the International Journal of Science and Research Archive (IJSRA), Vol. 12, No. 1, 2024 — DOI: 10.30574/ijrsra.2024.12.1.0781.

Portfolio V2 — Full-Stack Personal Brand Platform | [\[GitHub\]](#)

Stack: React 19 · Three.js · Vite · Firebase · Gemini AI · Framer Motion

- Built a production-deployed personal portfolio with an interactive Three.js 3D environment rendered in the browser, using custom WebGL shaders and Framer Motion animations to deliver a recruiter-engaging first impression.
- Integrated Google Gemini AI as a conversational assistant trained on a custom SHAGUN_CONTEXT system prompt — allowing visitors to ask questions about experience, projects, and skills and receive accurate, contextual answers in real time.
- Engineered for SEO with structured JSON-LD schema, Open Graph and Twitter Card meta tags, and a Vite-optimized production build deployed on GitHub Pages with custom 404 handling for React Router deep links.
- Architected on Firebase Cloud Functions for the AI backend and Firestore for chat persistence, with full CI/CD via GitHub Actions on every push to main.

RESEARCH PUBLICATION

Health Monitoring System Using ECG and PPG Techniques

International Journal of Science and Research Archive (IJSRA) | Vol. 12, No. 1, 2024 | Shagun Tyagi et al.

DOI: <https://doi.org/10.30574/ijrsra.2024.12.1.0781>

EDUCATION

Master of Business Administration (MBA)

Chandigarh University | In Progress

Jan 2026 – Dec 2027 (Expected)

B.Tech, Computer Science Engineering

Meerut Institute of Engineering and Technology (MIET) | CGPA: 8.4 / 10

Jul 2020 – Dec 2024

CERTIFICATIONS

- AWS Cloud Practitioner — AWS Academy
- Microsoft Azure Fundamentals (AZ-900) — Microsoft
- Microsoft Power Platform Fundamentals (PL-900) — Microsoft
- Cisco CyberOps Associate — Cisco
- CCNA: Introduction to Networks — Cisco
- Cybersecurity Essentials — Cisco
- Python Certification — HackerRank
- SQL Certification — HackerRank
- TCS Youth Employment Program — TCS

ACHIEVEMENTS

- Winner — National Level Hackathon: Led team to build a Waste Management Portal (Django + MySQL).
- Runner-up — Health-A-Thon: Developed an AI-powered health solution in 48 hours, recognized for innovation.
- Runner-up — CSS Battleground (MIET): Ranked 2nd among 250 candidates in a competitive front-end coding contest.
- Led CXI Community ML Team: Organized 5+ technical workshops on ML, AI, and cloud computing.